

Bluetooth Low Energy (BLE), ESP32, NanoBeacon, Firebase, Employee Tracking, Web Dashboard, UUID, IoT (Internet of Things), Attendance Monitoring, Workforce Management

4.1 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

- **BLE (Bluetooth Low Energy):** A wireless communication technology designed for short-range communication with very low power consumption. Used in this project for beacon-based employee detection.
- **ESP32:** A low-cost, low-power microcontroller with Wi-Fi and BLE capabilities. It is used as the receiver to detect BLE signals from employee beacons.
- **UUID (Universally Unique Identifier):** A unique code transmitted by each BLE beacon to distinguish it from others.
- **Firebase:** A platform developed by Google for creating mobile and web applications. In this project, it is used as the backend database to store and process attendance data.
- **NanoBeacon (IN100):** A small, energy-efficient BLE transmitter used as an employee tag for broadcasting signals.
- **Web Dashboard:** A browser-accessible interface that displays employee work logs and analytics in real-time.
- **IoT (Internet of Things):** A network of physical devices connected via the internet, capable of collecting and exchanging data. This project is an IoT application.
- **Attendance Monitoring:** The process of automatically logging employee entry and exit times and tracking work duration.
- **Workforce Management:** Strategies and tools used to track, manage, and improve employee performance and productivity.